

MULTIMODAL ASSISTIVE VISION SYSTEM FOR THE VISUALLY IMPAIRED USING SPATIAL AND VISION-LANGUAGE ARCHITECTURES

Team Members:

1. Varsha Vinod — AM.SC.U4CSE24159
2. Ajalya T M — AM.SC.U4CSE24162
3. Mokshitha Yarlagadda — AM.SC.U4CSE24170
4. Kanduri Venkata Sai Mani Avinash — AM.SC.U4CSE24171

Course: 23CSE473Neural Networks and Deep Learning

Abstract :

Visual impairment poses significant challenges in navigating surroundings and accessing important visual information during daily activities. Although existing assistive technologies can detect obstacles, they often fail to provide a comprehensive understanding of the environment or interpret textual information. This project proposes a **Multimodal Assistive Vision System** that utilizes Deep Learning techniques to improve environmental awareness and assist visually impaired individuals in real time.

The proposed system integrates multiple vision-based tasks, including object detection, scene understanding, and text recognition, to capture meaningful information from the user's surroundings. A spatial vision model is employed to identify and localize nearby objects, while a Vision-Language model generates descriptive information about the scene. An Optical Character Recognition (OCR) module extracts text from signs, labels, and documents. The information obtained from these modules is combined and converted into natural speech using a Text-to-Speech (TTS) system, enabling users to receive intuitive audio guidance. The system is designed to operate efficiently on portable devices, providing accurate and low-latency assistance for real-world applications.

By integrating spatial perception, scene understanding, and text recognition into a unified multimodal framework, the proposed system aims to enhance independent navigation, environmental awareness, and accessibility for visually impaired individuals. This project demonstrates the potential of modern Deep Learning techniques in developing intelligent, practical, and user-friendly assistive technologies.

Keywords: Deep Learning, Multimodal Learning, Computer Vision, Vision-Language Models, Object Detection, Optical Character Recognition (OCR), Text-to-Speech (TTS), Assistive Technology.